Program : **Diploma in Chemical Engineering**	
Course Code : **3074**	Course Title: **Fluid Mechanics**
Semester : **3**	Credits: **4**
Course Category: **Program Core**	
Periods per week: **4 (L: 3 T: 1 P: 0)**	Periods per semester: **60**

Course Objectives:

- To develop an understanding about the properties of fluid and basic laws of fluid statics.
- To develop an understanding of fluid handling techniques.

Course Pre-requisites:

Topic	Course code	Course name	Semester
Knowledge of basic Mathematics		Mathematics I Mathematics II	1 2
Knowledge of basic Physics		Applied Physics I Applied Physics II	1 2

Course Outcomes :

On completion of the course student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Interpret the properties of fluid and the laws of fluid statics.	12	Applying
CO2	Apply the theories of fluid dynamics to compute the energy associated with moving fluids.	18	Applying
CO3	Summarize the constructional details and working of different type of fluid handling equipment.	18	Understanding
CO4	Interpret piping system and flow measurements.	12	Applying

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	2						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcome	Description	Duration (Hours)	Cognitive Level
CO1	**Interpret the properties of fluid and the laws of fluid statics.**		
M1.01	Outline the properties of Fluids	1	Understanding
M1.02	Interpret the classification fluids using Newton's law of viscosity	1	Understanding
M1.03	Interpret the laws of fluid statics	5	Applying
M1.04	Comprehend the working of pressure measuring instruments	5	Understanding
Contents: State the properties of fluids: Density, specific gravity, viscosity, viscosity index and surface tension. Newton's law of viscosity - Classification of fluids. Laws of fluid statics and its applications – Hydrostatic equilibrium – Solve simple problems based on hydrostatic equilibrium. Pressure - Gauge pressure - Absolute pressure - Total pressure - Units of pressure - Pressure measuring using manometers - U tube manometer - Inclined tube manometer-Properties of manometric fluid.			
CO2	**Apply the theories of fluid dynamics to compute the energy associated with moving fluids.**		
M2.01	Interpret the flow regimes in a moving fluid.	4	Applying
M2.02	Compute the total mechanical energy of a moving fluid.	5	Applying
M2.03	Estimate the head loss due to friction in fluids.	5	Applying

M 2.04	Estimate the losses due to expansion and contraction.	4	Applying
	Series Test – I		

Contents:

Continuity equation - rate of discharge- average velocity - mass flow rate - volumetric flow rate- simple problems - Reynolds number- Reynolds experiment - Critical velocity- velocity profile- viscous and turbulent flow- simple problems - Identify steady state and unsteady state flow - potential flow - State momentum balance

Bernoulli's equation- conservation of energy - correction for fluid friction - simple problems.

Head loss due to friction - Skin friction - Foam friction - Effect of roughness - Estimation of frictional losses in straight pipes - Fanning's friction factor - Darcy's coefficient - simple problems

Estimate the losses due to expansion and contraction - Equivalent length - Hydraulic radius - Equivalent diameter

CO3	**Summarize the constructional details and working of different type of fluid handling equipment.**		
M3.01	Summarise the transportation of liquids using centrifugal pump.	4	Understanding
M3.02	Summarise the transportation of liquids using positive displacement pump.	5	Understanding
M3.03	Summarise the transportation of solids using fluidization.	5	Understanding
M3.04	Illustrate the Vacuum Producing Equipments used in Industry	4	Understanding

Contents:

Centrifugal pump - multistage pump - Identify centrifugal pump parts– Impeller, casing, stuffing box, mechanical seal - Centrifugal pump working - Characteristics curves - Advantages and disadvantages of centrifugal pump - NPSH and cavitation.

Positive displacement pumps: Piston pumps and rotary pumps- Simplified diagram, working and application - Piston pump, plunger, diaphragm pump - Rotary pumps: gear pumps, vane pumps, screw pumps - Pulsation dampner and relief valves on positive displacement pump - Safe operation of centrifugal and positive displacement pumps - Comparison of centrifugal and positive displacement pumps.

Fluidization – Application of fluidization in transportation of solids – Advantages and applications of fluidization.

Vacuum producing equipments: ejectors- working, diagram and applications, vacuum pumps- reciprocating and liquid ring.

CO4	**Interpret piping system and flow measurements.**		
M4.01	Outline the Piping systems	2	Remembering
M4.02	Comprehend the functions of gaskets and packing in pipe fittings.	2	Understanding
M4.03	Illustrate the types of valves used in industries.	4	Understanding
M4.04	Interpret the working of flow measuring equipment.	4	Applying
	Series Test – II		

Contents:

Pipes and tubes: Define schedule number, Birmingham wire gauge, international standards of pipes and fittings API and ASME, piping materials, Color coding in industry

Gaskets and packing in pipe fittings: types and materials.

Application and uses of commonly used valves - Gate valve, Globe valve, Needle valve, Diaphragm valve, Butterfly valve, Ball valve, Plug valve, check valve – swing check, ball check, lift check valve.

Flow measuring devices, working principle, diagram, flow pattern, derivations, simple problems: Orifice meter, Ventury meter, advantages and disadvantages.
Working principle, diagram; Pitot tube, weir, notches, rotameter

Text / Reference

T/R	Book Title/Author
T1	Unit Operations; Warren.L.McCabe Julian.C.Smith
R1	Introduction to Chemical Engineering; Walter.L.Badger & Julius.T.Banchero
R2	Fluid Mechanics; Yunus A Cengel, John M Cimbala

Online Resources

Sl.No	Website Link
1	https://nptel.ac.in/courses/103/104/103104043/